

AUDIT

Localization + per-language SEO audit — generated chrome (both sites)

Scope: every **generator-emitted** user-visible string that is NOT part of the `_content/** docs`, plus per-language SEO on generated pages. Source of truth is the Go generator `_tools/gen/` (`shells.go`, `seo.go`, `home.go`, `portfolio.go`, `product.go`) and the site JS i18n dictionaries (`milosvasic.ru/assets/js/i18n.js` → `MV_I18N`; `vasic.digital/js/translations.js`). The `_content_<lang>/` doc-translation batch and `_tests/evidence/translate-new/` are a SEPARATE surface and are out of scope (only read, never written).

LANGS (15): en · ru sr de es fr be zh kk hi ja ko ar tr fa

1. Surfaces and how each is localized

Surface	Renderer	Localization mechanism	State BEFORE
milosvasic.ru home	Jekyll _layouts/default.html + home.go (jekyll)	client-side i18n.js (MV_I18N, 15 langs) + language switcher	GOOD (chrome localized)
vasic.digital home	home.go (standalone)	data-i18n attrs present, but translations.js not loaded → inert	EN-ONLY in practice
Portfolio (both sites)	portfolio.go product.go	none — chrome hard-coded English	EN-ONLY, all 15

Surface	Renderer	Localization mechanism	State BEFORE
Product pages (both sites)		none — chrome hard-coded English	EN-ONLY, all 15

Key finding: the **server-rendered** portfolio + product pages emitted 100% hard-coded English chrome and English-only SEO **for every language**, and no per-language product/portfolio pages had ever been generated on disk (products/<lang>/, portfolio/<lang>/ were absent).

2. Inventory of generated non-content UI strings (the gap set)

These are the strings the generator prints. BEFORE = hard-coded English for all 14 non-EN languages (English-fallback). Each now has a key in ui-i18n.json.

Chrome (shells / product / portfolio): - skip “Skip to content” (skip link) - nav.home “Home”, nav.products “Products”, nav.portfolio “Portfolio” - toggle “Toggle theme” (theme-toggle aria-label) - footer.suffix “built on the OpenDesign system”

Portfolio body: - pf.eyebrow “unified portfolio”, pf.title “Portfolio” - pf.download “Download Portfolio (PDF)” - pf.stat.products “Products”, pf.stat.tiers “Tiers”, pf.stat.helix “Helix products” - pf.overview.eyebrow “overview”, pf.overview.title “One fleet, one discipline” - pf.readmore “Read more” - pf.lede (hero lede paragraph), pf.summary (overview paragraph) - tier1/2/3.eyebrow + tier1/2/3.title (six tier-section headers)

Product body: - prod.tier “tier”, prod.order “order” (eyebrow “// tier: X · order N”) - prod.source “Source”, prod.license “license” - prod.architecture “architecture” (diagram figcaption)

SEO: - kw.vasic / kw.milos (per-site meta keywords — previously ABSENT)

Total: 29 distinct generator UI/SEO keys × 14 non-EN languages = **406 strings** that were English-fallback before this work.

3. Per-language SEO audit of generated pages (BEFORE)

Signal	Before	Notes
<title>	English only	product = “ — ”; portfolio = “Portfolio — ”
meta description	English only	product = EN summary; portfolio = EN lede
meta keywords	ABSENT	no keywords tag emitted at all
link canonical	OK	correct EN canonical per page
hreflang alternates	OK (reciprocal)	full 15-lang matrix + x-default; reciprocal by construction
og:* / twitter:*	present, EN	title/desc English; og:locale hard-coded en_US for every language
JSON-LD	EN	WebSite.inLanguage = “en”, SoftwareApplication.inLanguage = “en” (hard-coded)

So: hreflang + canonical were already correct; **title/description were not translated, keywords were missing entirely, and og:locale + JSON-LD inLanguage were hard-coded English on every localized page.**

4. JS i18n dictionary audit

- MV_I18N (milosvasic.ru/assets/js/i18n.js): 15 langs, comprehensive home keys (nav, hero, downloads, summary, work cards, skills, author, contact). Reviewer-gated per project history. GOOD.
- translations.js (vasic.digital/js/translations.js): 15 langs present, BUT the keys are **stale** relative to the regenerated vasic home (e.g. it has hero.title = “Excellence in Software Engineering” while the generated page uses hero.title = “AI-native software engineering, built to be trusted.” and keys like hero.eyebrow, hero.cta_products, stat.providers, about.body2 that translations.js does not carry). It is ALSO not loaded by the generated index.html. Net effect: the vasic home language switcher is non-functional and, if wired as-is, would mistranslate. (Documented gap — see §6.)
- Untranslated aria-labels on the milosvasic Jekyll layout (“Change language”, “Toggle dark / light theme”, “Close”,

“Language”, “Theme”) and the footer prose line have no data-i18n. (Minor; layout-owned, documented gap.)

5. FILL — what changed in the generator (this work)

- New `_tools/gen/i18n.go` + embedded `ui-i18n.json`: a per-language dictionary (`T(lang, key)` with explicit EN fallback), `htmlDir` (RTL for `ar/fa/he/ur`), `ogLocale`, `siteKeywords`. Reviewer-gated translations (see §7).
- `seo.go`: `og:locale` now per-language; meta keywords emitted per-language; JSON-LD `inLanguage` per-language via `baseGraphLang` / `softwareApplicationNodeLang` (original signatures preserved → non-breaking). `pageSEO` gained `lang` + keywords.
- `portfolio.go`: hero eyebrow/title/lede, download CTA, three stat labels, overview eyebrow/title, summary, all six tier headers, per-card “Read more”, skip link, nav (Home/Portfolio), theme-toggle aria, footer suffix, `<html lang/dir>`, and the SEO head (title/desc/keywords/og:locale/inLanguage) — all localized via `T(lang, ...)` for both the self-contained and Jekyll shells.
- `product.go`: eyebrow (“// tier: ... · order ...”), “Source”, “license”, diagram “architecture” caption, skip link, nav (Home/Products), theme-toggle aria, footer suffix, `<html lang/dir>`, and SEO head — all localized. Product BODY is read-only sourced from `_content_<lang>/` when a translation exists (e.g. `ru`), else `EN` (that doc surface is owned elsewhere).
- `main.go`: threads `lang` into `renderProduct`.

Glossary terms (Helix*, LLM, Vasic Digital, Server Factory, PDF, Go, Kotlin, ...) are preserved verbatim in every translation via `glossary_protect.py` sentinels.

6. Honest caveats / residual gaps (not closed here)

1. **vasic.digital home client i18n**: `translations.js` is stale + unloaded. The generated vasic home remains EN-only until `translations.js` is regenerated against the current `data-i18n` keys (a client-content task) and wired into the generated `index.html`. Server-rendered vasic home chrome now uses `T()` but is only ever generated for `en` (home content data is EN-only), so localization of the vasic home still depends on the client switcher. NOT closed.
2. **milosvasic home**: chrome is already localized client-side; the remaining gaps are a few layout aria-labels + footer prose

without data-i18n, and jekyll-seo-tag head being English-only.
Layout-owned; NOT closed here.

3. **Product/portfolio card blurbs + product body** are
_content-derived (taglines/summaries). Chrome + SEO around
them is localized; the prose is the separate _content_<lang>
surface (used read-only where present).
4. TestAvailableLangs currently fails in this checkout because
_content_de/ and _content_sr/ are not on disk this session (only
_content_ru/). This is environmental, predates this work, and is
not touched (editing _content_* is out of scope).

See coverage-matrix.md for the before→after numbers and reviews/
for the independent per-language review verdicts.